

Part (a) Solution:

Average Requests/Sec = 20

Avg Size = 500 KB * 8 = 4000 Kbits = 4 Mbits

Total load = $(4.0 * 20) = 80$ Mbits/sec

LAN Utilization = 80 %

Data rate to browsers over access link = $(0.33 * 80) = 26.4$ Mbits

Cache hit ratio = 0.33 which would put load of 26.4 Mbits/sec to access link

Access link utilization = $26.4/15 = 1.76$

So, it will be 176% utilized and packets will drop due to queuing.

Part (b)

i) Solution:

It takes 9RTT for the first client to obtain the IP address for google.com

ii) Solution:

It takes RTT for the second client (as the IP address is already in local domain server)

Part (c) Solution:

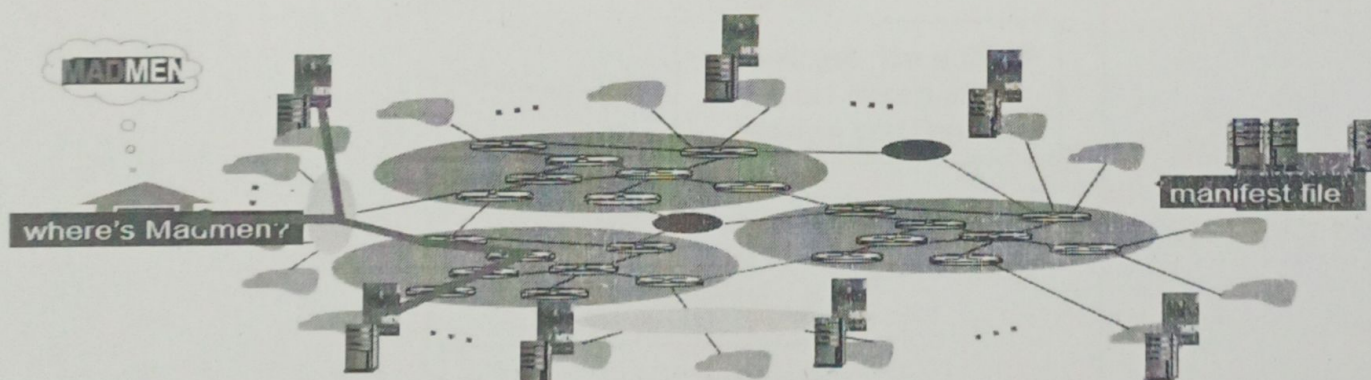
This is similar to the file upload to multiple peer we studied in the class.

Client-server model needs more computer, storage and network bandwidth as the number of users accessing the shared file folders grows to a large number (video lecture where size is ~1000s of MB). A P2P based application on each peer (like Bit torrent running on LAN) will allow a new user to use multiple copies of lectures those stored on different peers (as well as server copy) to take part of the download. Each peer PC will provide the bandwidth hence removing the bottleneck.

Part (d) Solution:

Content Distribution Networks (CDNs)

- CDN: stores copies of content at CDN nodes
 - e.g. Netflix stores copies of MadMen
- subscriber requests content from CDN
 - directed to nearby copy, retrieves content
 - may choose different copy if network path congested



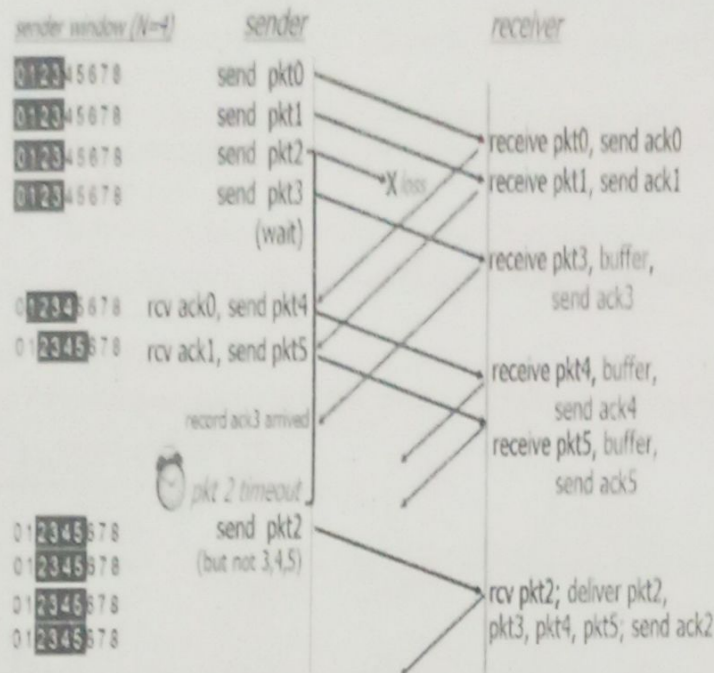
Question # 2:

Part (a) Solution:

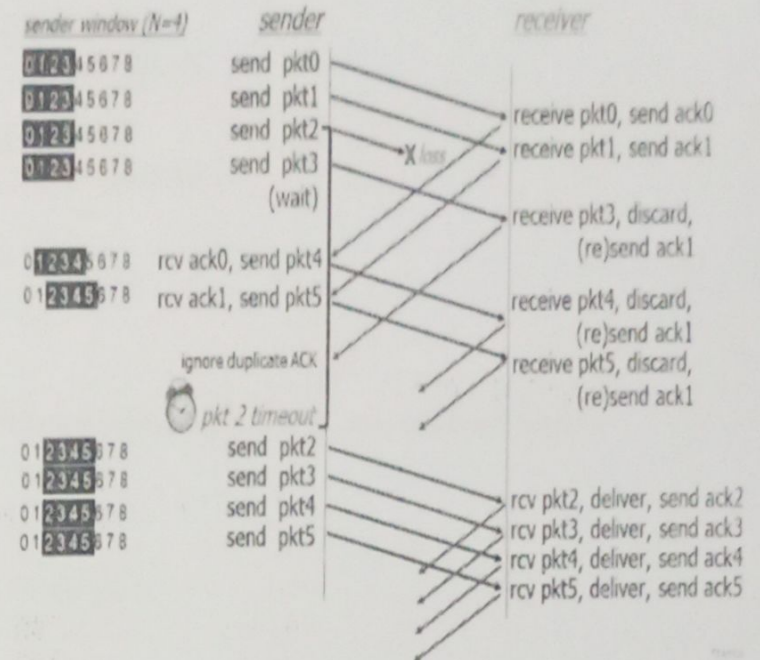
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[2 * 10 = 20 Points]

Selective Repeat in action



Go-Back-N in action



Part (b)

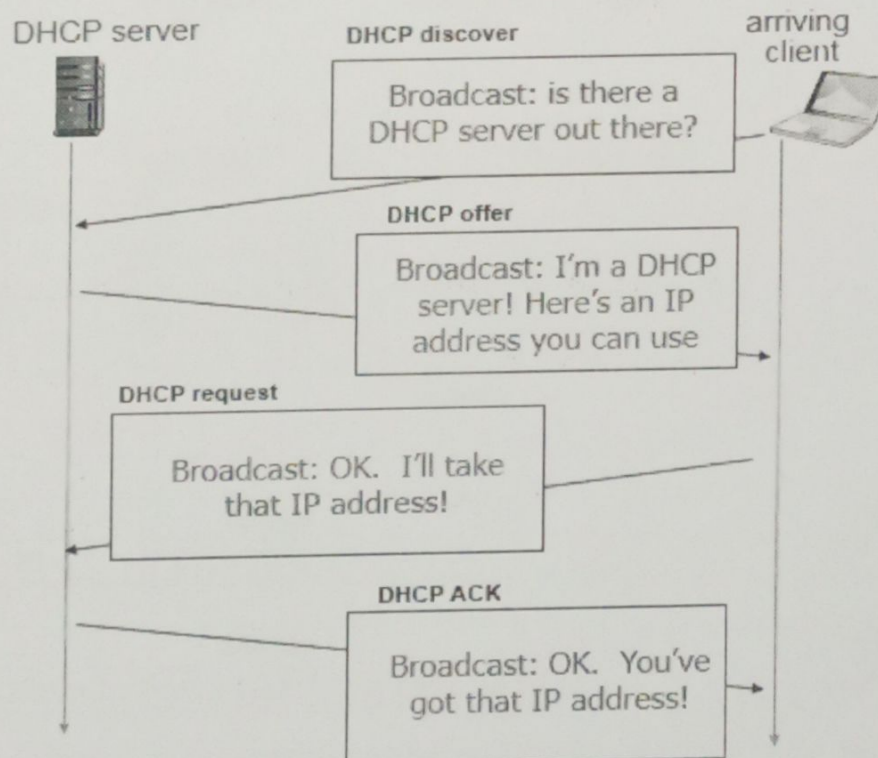
- Solution: Triple Duplicate Ack
- Solution: No, it could be due to reordering. Due to queuing and asymmetric paths.
- Solution: Timeout
- Solution: No. Congestion in either direction could cause $RTT > RTO$ (retransmission timeout)
- Solution: Slow Start is operating during event A.

Question # 3:

Part (a) Solution:

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[05 * 04 = 20 Points]



Part (b) Solution:

IP address, IP address of first hop router, name and IP address of DNS server, Network subnet mask.

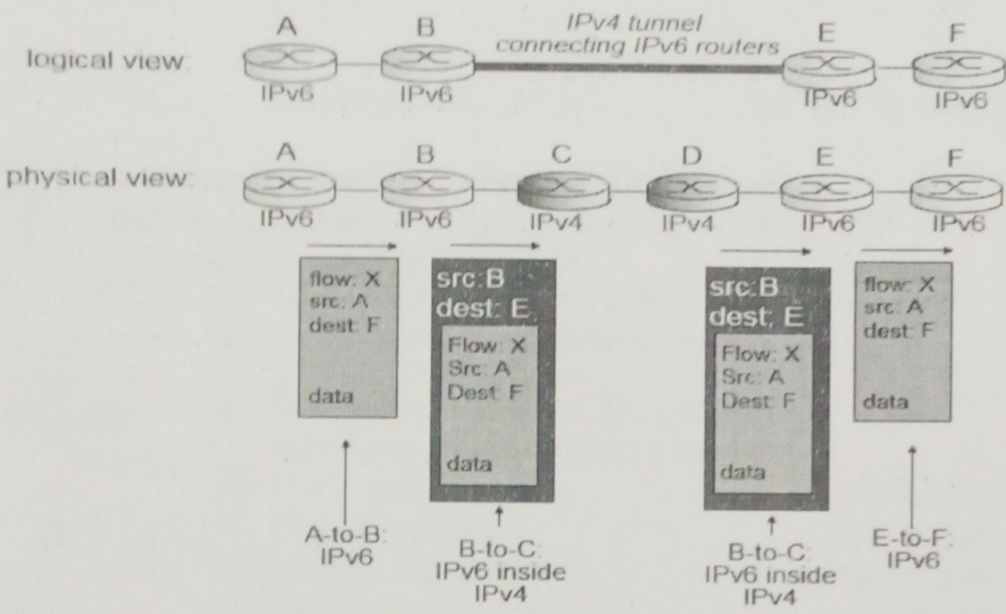
Part (c)

- i) Solution:
The source address will be the local host's IP, which is 10.0.1.11
The destination address will be the remote machine's IP, which is 128.119.176.181
- ii) Solution:
The source address will be the router's public IP, which is 135.122.200.207
The destination address will be the remote machine's IP, which is 128.119.176.181
- iii) Solution:
The source address will be the remote machine's IP, which is 128.119.176.
The destination address will be the 181router's public IP, which is 135.122.200.207
- iv) Solution:
The source address will be remote machine's IP, which is 128.119.176.181
The destination address will be the local host's IP, which is 10.0.1.11

Part (d) Solution:

It is implemented using tunneling. IPv6 datagram carried as payload in IPv4 datagram among IPv4 routers as shown below.

Tunneling



Part (e) Solution:

	Total Length	Flag	Fragment offset
Original Packet	3980		
Fragment 1	1500	1	0
Fragment 2	1500	1	185
Fragment 3	1020	0	370

Question # 4:

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[08 * 05 = 40 Points]

Part (a) Solution:

(a)	Match	Action
	IP Src = 10.3.*.* ; IP Dst = 10.2.*.*	Forward(3)
(b)	Match	Action
	Ingress Port = 1 ; IP Src = 10.3.*.* ; IP Dst = 10.2.*.*	Forward(4)
(c)	Match	Action
	Ingress port = 2 ; IP Dst = 10.2.0.3	Forward(3)
	Ingress port = 2 ; IP Dst = 10.2.0.4	Forward(4)

Part (b) Solution:

The Department-1 LAN-1 will require 50 host IP addresses.

Determine the following.

- a) Number of Bit Borrowed= 2
- b) Total Number of Subnets= $2^2 = 4$
- c) Total Number of Host Addresses= 64
- d) Number of Usable Addresses= 62
- e) Custom Subnet Mask= 255.255.255.192

Subnet Range (Network & Broadcast address)

192.168.9.0 ---192.168.9.63

192.168.9.128---192.168.9.191

192.168.9.64 ---192.168.9.127

192.168.9.192 ---192.168.9.255

The Department-1 LAN-2 will require 120 host IP addresses.

Determine the following.

- a) Number of Bit Borrowed= 1
- b) Total Number of Subnets= $2^1 = 2$
- c) Total Number of Host Addresses= 128
- d) Number of Usable Addresses= 126
- e) Custom Subnet Mask= 255.255.255.128

Subnet Range (Network & Broadcast address)

192.168.9.0 ---192.168.9.127

192.168.9.128 ---192.168.9.255

The Department-2 LAN 1 will require 10 host IP addresses.

Determine the following.

- a) Number of Bit Borrowed= 4
- b) Total Number of Subnets= $2^4 = 16$
- c) Total Number of Host Addresses= 16
- d) Number of Usable Addresses= 14
- e) Custom Subnet Mask= 255.255.255.240

Subnet Range (Network & Broadcast address)

192.168.9.0 ---192.168.9.15

192.168.9.32---192.168.9.47

192.168.9.64 ---192.168.9.79

192.168.9.96 ---192.168.9.111

192.168.9.128 ---192.168.9.143

192.168.9.160 ---192.168.9.175

192.168.9.192 ---192.168.9.207

192.168.9.224 ---192.168.9.239

192.168.9.16 ---192.168.9.31

192.168.9.48 ---192.168.9.63

192.168.9.80 ---192.168.9.95

192.168.9.112 ---192.168.9.127

192.168.9.144 ---192.168.9.159

192.168.9.176 ---192.168.9.191

192.168.9.208 ---192.168.9.223

192.168.9.240 ---192.168.9.255

The Department-2 LAN 2 will require 18 host IP addresses.

Determine the following.

- a) Number of Bit Borrowed= 3
- b) Total Number of Subnets= $2^3 = 8$
- c) Total Number of Host Addresses= 32
- d) Number of Usable Addresses= 30
- e) Custom Subnet Mask= 255.255.255.224

Subnet Range (Network & Broadcast address)

192.168.9.0 ---192.168.9.31

192.168.9.64---192.168.9.95

192.168.9.128 ---192.168.9.159

192.168.9.192 ---192.168.9.223

192.168.9.32 ---192.168.9.63

192.168.9.96 ---192.168.9.127

192.168.9.160 ---192.168.9.191

192.168.9.224---192.168.9.255

• **Part (c) Solution:**

Steps	Node (N')	D(x), p(x)	D(u), p(u)	D(v), p(v)	D(t), p(t)	D(y), p(y)	D(z), p(z)
0	w	6,w	3,w	4,w	∞	∞	∞
1	wu	6,w		4,w	5,u	∞	∞
2	wuv	6,w			5,u	12,v	∞
3	wuvt	6,w				12,v	∞
4	wuvtx					12,v	14,x
5	wuvtxy						14,x
6	wuvtxyz						

Forwarding Table	
t	(w, u)
u	(w, u)
v	(w, v)
x	(w, x)
y	(w, v)
z	(w, x)

Part (d)

i) Solution:

eBGP: 3a,4a,4c,2a,1b,1a

iBGP: 3c,3b,4b,2c,2b,1d,1c,1d

ii)Solution:

AS3, AS4, AS2, AS1, B OR 3b, 3a, 4a, 4b, 4c, 2a, 1b, 1d, B

Part (e)

i) Solution:

PC-B will send an ARP reply with its MAC address (CC-49-DE-D0-AB-7D).

ii) Solution:

It will send an ARP request for the MAC address of the default gateway (R) i-e (E6-E9-00-17-BB-4B)

iii) Solution:

The IP address of PC-C (222.222.222.222)

Best of Luck!!!